

TRANSPORTATION PROBLEM

PROBLEM:

A mining company extracts gravel, the basic product it sells, from three mines, L_1 , L_2 , and L_3 . The weekly production of each mines is 75, 150 and 75 tones of gravel respectively. The gravel has to be transported to five main consumers, K_1 , K_2 , K_3 , K_4 and K_5 requiring for their needs 100, 60, 40, 75 and 25 tones of gravel per week respectively. The problem that concerns the company's management is the minimization of the required cost for the transportation of the product to the consumers. For this purpose a detailed cost analysis was carried out which gave the results of the following table (the numbers denote the transportation cost in £ per ton of gravel).

		Consumers				
Mines		K_1	K_2	K_3	K_4	K_5
	L_1	3	2	3	4	1
	L_2	4	1	2	4	2
	L_3	1	0	5	3	2

AIM:

To find an optimal transportation plan that minimizes the total cost.

CODE IN R LANGUAGE:

```
library(lpSolve)
cost.mat=matrix(nrow = 3, ncol = 5)
cost.mat[1,]=c(3, 2, 3, 4, 1)
cost.mat[2,]=c(4, 1, 2, 4, 2)
cost.mat[3,]=c(1, 0, 5, 3, 2)
direction='min'
row.signs=rep('<=', 3)
row.rhs=c(75, 150, 75)
col.signs=rep('>=', 5)
col.rhs=c(100, 60, 40, 75, 25)
solution=lp.transport(cost.mat, direction, row.signs, row.rhs, col.signs,
col.rhs)
solution$solution
```

OUTPUT:

Success: the objective function is 615 [1] [2] [3] [4] [5]

[1,]	25	0	0	25	25
[2,]	0	60	40	50	0
[3,]	75	0	0	0	0

RESULT:

Mine L_1 transports 25 tons of gravel to consumers K_1 , K_4 , and K_5 .

Mine L_2 transports 60 tons of gravel to consumer K_2 , 40 tons to consumer K_3 , and 50 tons to consumer K_4 .

Mine L_3 transports 75 tons of gravel to consumer K_1 .

The final minimized cost is 615.

BOOTSTRAPPING

PROBLEM:

Write a R program to perform bootstrapping on the Auto dataset and evaluate the model.

AIM:

Perform bootstrapping on the given set of data

CODE IN R LANGUAGE:

```
library(MASS)
attach(Auto)
statistic=function(data, index) {
  lm.fit=lm(mpg ~ horsepower, data = data, subset = index)
  coef(lm.fit)
}
statistic(Auto, 1:392)
summary(lm(mpg ~ horsepower, data = Auto))
set.seed(123)
boot_linear=boot(Auto, statistic, 1000)
quad_statistic <- function(data, index) {
  lm.fit <- lm(mpg ~ poly(horsepower, 2), data = data, subset = index)
  coef(lm.fit)
}
set.seed(1)
boot_quad=boot(Auto, quad_statistic, 1000)
boot_quad
summary_lm_quad=summary(lm(mpg ~ poly(horsepower, 2), data =
Auto))
summary_lm_quad
```

OUTPUT:

```
>library(MASS)
> attach(Auto)
> statistic=function(data, index) {
+   lm.fit=lm(mpg ~ horsepower, data = data, subset = index)
+   coef(lm.fit)
+ }
> statistic(Auto, 1:392)
(Intercept) horsepower
39.9358610 -0.1578447
> summary(lm(mpg ~ horsepower, data = Auto))
Call:
```

```
lm(formula = mpg ~ horsepower, data = Auto)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.5710	-3.2592	-0.3435	2.7630	16.9240

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	39.935861	0.717499	55.66	<2e-16 ***
horsepower	-0.157845	0.006446	-24.49	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.906 on 390 degrees of freedom

Multiple R-squared: 0.6059, Adjusted R-squared: 0.6049

F-statistic: 599.7 on 1 and 390 DF, p-value: < 2.2e-16

boot(data = Auto, statistic = quad_statistic, R = 1000)

ORDINARY NONPARAMETRIC BOOTSTRAP

Bootstrap Statistics :

	original	bias	std. error
t1*	23.44592	-0.003660358	0.2195369
t2*	-120.13774	0.002769239	3.6138046
t3*	44.08953	0.101767465	4.1998076

```
> summary_lm_quad=summary(lm(mpg ~ poly(horsepower, 2), data = A  
uto))
```

```
> summary_lm_quad
```

Call:

```
lm(formula = mpg ~ poly(horsepower, 2), data = Auto)
```

Residuals:

Min	1Q	Median	3Q	Max
-14.7135	-2.5943	-0.0859	2.2868	15.8961

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	23.4459	0.2209	106.13	<2e-16 ***
poly(horsepower, 2)1	-120.1377	4.3739	-27.47	<2e-16 ***
poly(horsepower, 2)2	44.0895	4.3739	10.08	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.374 on 389 degrees of freedom

Multiple R-squared: 0.6876, Adjusted R-squared: 0.686

F-statistic: 428 on 2 and 389 DF, p-value: $< 2.2e-16$

RESULT:

A regression model for the in-built dataset Auto is built using bootstrapping method to predict the variable 'mpg'